KAIZEN MATHS

Lesson starter

Confidence Intervals

Confidence Interval for a Proportion

Name

S1

Calculate the confidence interval for a population proportion.

In a sample of 150 people, 39 support a proposal. Find a 95% confidence interval for the population proportion who support it.

S2

Use the survey data to estimate a population proportion.

A school survey found that 187 out of 400 students use an online revision tool weekly. Find a 90% confidence interval for the true proportion.

S3

Interpret the proportion interval carefully.

A 99% confidence interval for a population proportion is (0.199, 0.311). Interpret the interval.

S4

Use the interval to decide whether a claimed proportion is plausible.

A 90% confidence interval for a population proportion is (0.286, 0.403). A claim says the true proportion is 0.7. Is this claim plausible at this confidence level?

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KAIZEN MATHS

Exit ticket

Confidence Intervals

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EA1

Calculate the confidence interval for a population proportion.

In a sample of 120 people, 73 support a proposal. Find a 90% confidence interval for the population proportion who support it.

EA2

Use the survey data to estimate a population proportion.

A school survey found that 116 out of 180 students use an online revision tool weekly. Find a 99% confidence interval for the true proportion.

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Exit ticket

Confidence Intervals

Confidence Interval for a Proportion

Name

EB1

Interpret the proportion interval carefully.

A 99% confidence interval for a population proportion is (0.180, 0.367). Interpret the interval.

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Confidence Intervals

Confidence Interval for a Proportion

Name

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A 99% confidence interval for a population proportion is (0.180, 0.367). Interpret the interval.

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Teacher answers & working

Confidence Intervals · Confidence Interval for a Proportion

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S1

Calculate the confidence interval for a population proportion.

In a sample of 150 people, 39 support a proposal. Find a 95% confidence interval for the population proportion who support it.

Answer:

95% CI: (0.1898, 0.3302).

Working:

Estimate the population proportion using $\hat{p} = \frac{39}{150} = 0.2600$.

Use $\hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

For 95% confidence, z = 1.960.

Standard error = $\sqrt{\frac{0.2600(1-0.2600)}{150}} = 0.03581$.

Margin of error = 1.960 × 0.03581 = 0.07020.

Interval = 0.2600 ± 0.07020 = (0.1898, 0.3302).

S2

Use the survey data to estimate a population proportion.

A school survey found that 187 out of 400 students use an online revision tool weekly. Find a 90% confidence interval for the true proportion.

Answer:

90% CI: (0.4265, 0.5085).

Working:

$\hat{p} = \frac{187}{400} = 0.4675$.

Standard error = $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

Standard error = $\sqrt{\frac{0.4675(1-0.4675)}{400}} = 0.02495$.

For 90% confidence, z = 1.645.

Margin of error = 1.645 × 0.02495 = 0.04104.

Confidence interval = 0.4675 ± 0.04104 = (0.4265, 0.5085).

S3

Interpret the proportion interval carefully.

A 99% confidence interval for a population proportion is (0.199, 0.311). Interpret the interval.

Answer:

We are 99% confident that the true population proportion lies between 0.199 and 0.311.

Working:

This interval is about the true population proportion.

The sample proportion is only the estimate used to build the interval.

Interpretation: We are 99% confident that the true population proportion lies between 0.199 and 0.311.

Teacher answers & working

Confidence Intervals · Confidence Interval for a Proportion

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S4

Use the interval to decide whether a claimed proportion is plausible.

A 90% confidence interval for a population proportion is (0.286, 0.403). A claim says the true proportion is 0.7. Is this claim plausible at this confidence level?

Answer:

The claim is not supported because it lies outside the interval.

Working:

Compare the claimed value 0.7 with the interval (0.286, 0.403).

0.7 is not between 0.286 and 0.403.

A value outside the interval is not supported by this confidence interval.

The claim is not supported because it lies outside the interval.

EA1

Calculate the confidence interval for a population proportion.

In a sample of 120 people, 73 support a proposal. Find a 90% confidence interval for the population proportion who support it.

Answer:

90% CI: (0.5350, 0.6816).

Working:

Estimate the population proportion using $\hat{p} = \frac{73}{120} = 0.6083$.

Use $\hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

For 90% confidence, z = 1.645.

Standard error = $\sqrt{\frac{0.6083(1-0.6083)}{120}} = 0.04456$.

Margin of error = 1.645 × 0.04456 = 0.07330.

Interval = 0.6083 ± 0.07330 = (0.5350, 0.6816).

Teacher answers & working

Confidence Intervals · Confidence Interval for a Proportion

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EA2

Use the survey data to estimate a population proportion.

A school survey found that 116 out of 180 students use an online revision tool weekly. Find a 99% confidence interval for the true proportion.

Answer:

99% CI: (0.5525, 0.7364).

Working:

$\hat{p} = \frac{116}{180} = 0.6444$.

Standard error = $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

Standard error = $\sqrt{\frac{0.6444(1-0.6444)}{180}}$ = 0.03568.

For 99% confidence, z = 2.576.

Margin of error = 2.576 × 0.03568 = 0.09191.

Confidence interval = 0.6444 ± 0.09191 = (0.5525, 0.7364).

EB1

Interpret the proportion interval carefully.

A 99% confidence interval for a population proportion is (0.180, 0.367). Interpret the interval.

Answer:

We are 99% confident that the true population proportion lies between 0.180 and 0.367.

Working:

This interval is about the true population proportion.

The sample proportion is only the estimate used to build the interval.

Interpretation: We are 99% confident that the true population proportion lies between 0.180 and 0.367.